ASSIGNMENT 1

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Subject Name: System Design
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Q1 : Explain SRP and OCP in detail with proper examples.
Solution:

1. Single Responsibility Principle (SRP)
Definition
A class should have only one reason to change.
This means **one class = one responsibility**.
Why SRP is important

- Easier maintenance
- Better readability
- Reduced impact of changes
- Improved testability

Example (Violation)

```
class Report {
   void generateReport() { }
   void saveToFile() { }
   void printReport() { }
}
```

One class handling **business logic + storage + printing**

SRP-Compliant Version

```
class ReportGenerator {
   void generateReport() { }
}
```

```
class ReportSaver {
   void saveToFile() { }
}

class ReportPrinter {
   void printReport() { }
}
```

Each class has **one responsibility**

2. Open–Closed Principle (OCP)

Definition

Software entities should be open for extension but closed for modification.

Why OCP is important

- Avoids breaking existing code
- Supports scalability
- Improves reliability

Example (Violation)

```
class Discount {
   double calculate(String type) {
      if(type.equals("Student")) return 0.1;
      if(type.equals("Senior")) return 0.2;
      return 0;
   }
}
```

Adding a new discount requires modifying the class

OCP-Compliant Version

```
interface Discount {
   double calculate();
}

class StudentDiscount implements Discount {
   public double calculate() { return 0.1; }
}
```

```
class SeniorDiscount implements Discount {
   public double calculate() { return 0.2; }
}
```

New discounts can be added without changing existing code

Q2. Discuss in detail about the violations in SRP and OCP along with their fixes.
Solution :

SRP Violation

Violation

- A class performs multiple tasks
- Example: Authentication + Logging + Database operations

Problems

- Difficult to maintain
- High coupling
- Hard to test

Fix

- Split responsibilities into separate classes

OCP Violation

Violation

- Use of if–else or switch for behavior changes
- Modifying existing code for new functionality

Problems

- Risk of introducing bugs
- Code becomes rigid

Fix

- Use interfaces, inheritance, and polymorphism

Q3. Design an HLD for an Online Examination System applying these principles.

Functional Requirements:

- User should be able to register and login
- System should support different roles (Student, Admin)

- Admin should be able to create and schedule exams
- Admin should be able to add/manage questions
- Student should be able to attempt exams online
- System should evaluate answers automatically
- System should generate results
- System should notify users about exam and results

Non Functional Requirements:

A. Scalability
- System should handle multiple students simultaneously during exams

B. Security
- Secure authentication and authorization
- Exam data should be protected

C. Performance
- Low response time during exam submission

D. Reliability
- No data loss during exam submission

E. Maintainability
- Easy to add new exam or evaluation types

Core Entities:

- User
- Exam
- Question
- Answer
- Result

API Endpoints

Auth & User Service APIs

- POST /login
- POST /register
- GET /users/{id}

Exam Management Service APIs

- POST /exams

- GET /exams

Question Bank Service APIs

- POST /questions
- GET /questions/{examId}

Evaluation Service APIs

- POST /evaluate

E. Result Service APIs

- GET /results/{userId}
-

High Level Diagram :

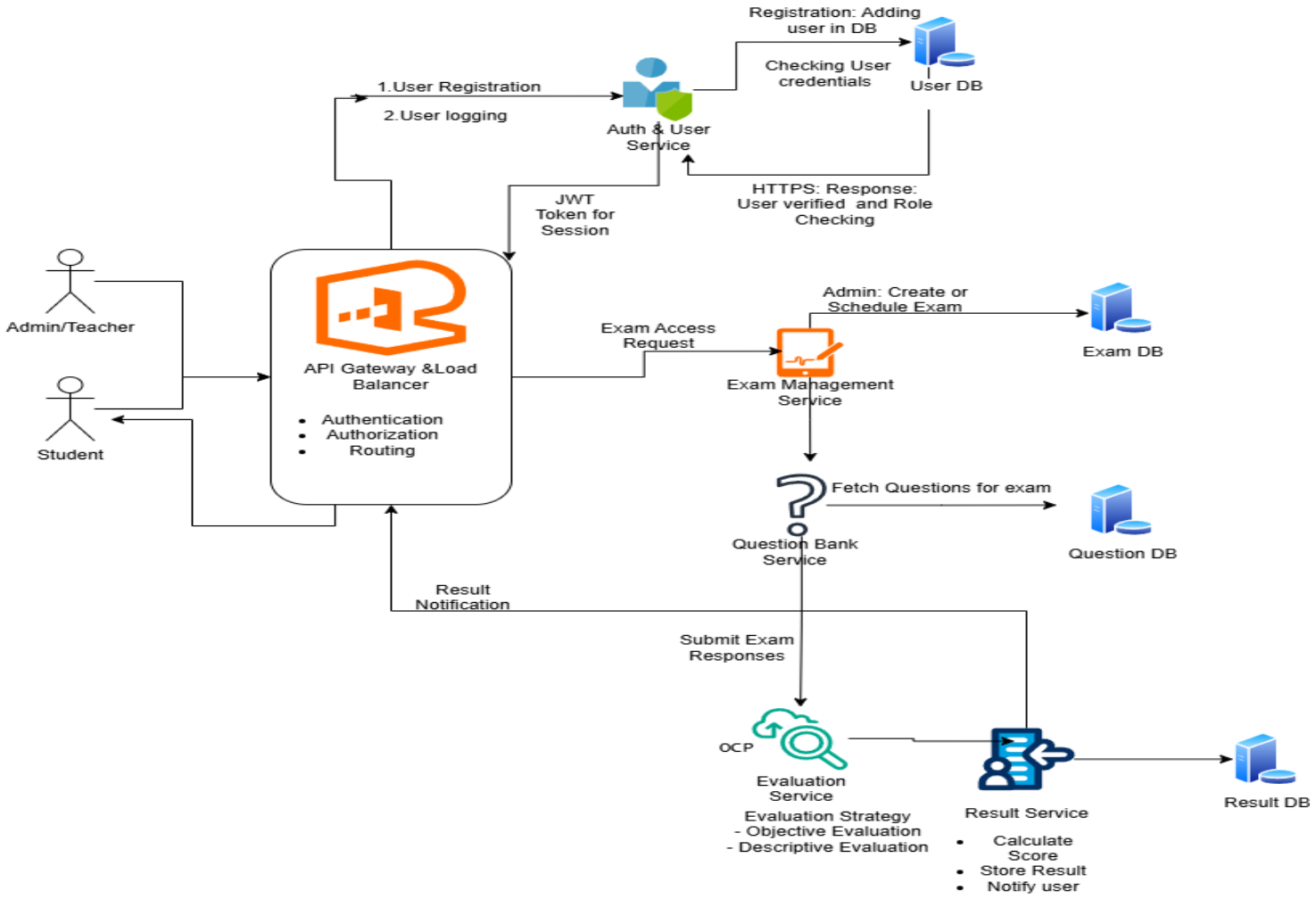

Figure 1: HLD of Online Examination System